Title

- Securing Multi-Cloud Environment: An Automated Data Deletion System with Integrated Intrusion Detection System Over Multi-Cloud Platforms

Problem Statement

- Cloud data can be exposed after unauthorized access.
- Organizations need automated response mechanisms.

Proposed Solution

- In this research, we proposed a development of an exhaustive system for automated data deletion in a multi-cloud environment with the capabilities of Intrusion detection. The proposed system aims to address the complex challenges associated with managing data privacy and security across multi-cloud infrastructures. The key components of the system include a policy driven data deletion mechanism, an intrusion detection system tailored for multi-cloud architectures, and a centralized policy engine for defining, evaluating, and enforcing policies governing data deletion, access control, and compliance.
- Our approach leverages policy-based management techniques to enable dynamic and flexible enforcement of data deletion policies, ensuring adherence to regulatory requirements and organizational policies. By integrating intrusion detection capabilities, the system enhances security posture and enables proactive identification and response to security threats and anomalies. Through this research, the aim is to contribute to the advancement of data management practices in multi-cloud environments and provide organizations with effective tools and methodologies for ensuring data privacy, compliance, and security in the cloud.

Key Concepts

- Cloud Computing
- Data Security
- Intrusion Detection
- Automated Response
- Cyber Security
- Cloud Security
- Research

Outcome

- The proposed automated data deletion system with the integration of intrusion detection system in a multi-cloud environment is theoretically validated through its proper calibration with already established data deletion mechanism, intrusion detection systems, security and compliance frameworks, data lifecycle management principles, automated response systems, fault tolerance theory, and compliance models. This provides foundation guarantees that the proposed system is designed to be secure, compliant, and efficient, providing a robust solution for protecting confidential and sensitive data in multi-cloud environments.